

HDOC DAY : 25

AIM: To write a C program that displays a menu to print different decreasing patterns using a switch statement and nested loops without using arrays. Each pattern is implemented in a separate parameterless function with no return value

ALGORITHM:

1. Start the program.
2. Display a menu with options for 4 different decreasing patterns and an exit option.
3. Read the user's choice using switch-case.
4. Ask the user to enter the number of rows (n).
5. Call the respective function based on the user's choice:
 - a. **Choice 1 (pattern1):** Use nested loops to print decreasing right-angled triangle using asterisks (*).
 - b. **Choice 2 (pattern2):** Use nested loops to print natural numbers from 1 to the remaining count in each row.
 - c. **Choice 3 (pattern3):** Use nested loops to print the row count repeated as many times as its value.
 - d. **Choice 4 (pattern4):** Use nested loops to print numbers in descending order starting from the remaining count down to 1.
6. Display the selected pattern.
7. End the program.

C CODE :

```
#include <stdio.h>
```

```
// Function for Pattern 1: Decreasing right-angled triangle
```

```
void pattern1() {
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for(i = n; i >= 1; i--) {
```

```
        for(j = 1; j <= i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
// Function for Pattern 2: Decreasing natural-number pattern
```

```
void pattern2() {
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
scanf("%d", &n);

for(i = n; i >= 1; i--) {
    for(j = 1; j <= i; j++) {
        printf("%d", j);
    }
    printf("\n");
}

// Function for Pattern 3: Decreasing repeated-number pattern
void pattern3() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {
        for(j = 1; j <= i; j++) {
            printf("%d", i);
        }
        printf("\n");
    }
}
```

```
// Function for Pattern 4: Reverse-number triangle
```

```
void pattern4() {
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for(i = n; i >= 1; i--) {
```

```
        for(j = i; j >= 1; j--) {
```

```
            printf("%d", j);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
}
```

```
int main() {
```

```
    int choice;
```

```
    printf("--- MENU ---\n");
```

```
    printf("1. Decreasing right-angled triangle\n");
```

```
    printf("2. Decreasing natural-number pattern\n");
```

```
    printf("3. Decreasing repeated-number pattern\n");
```

```
    printf("4. Reverse-number triangle\n");
```

```
    printf("Enter your choice (1-4): ");
```

```
scanf("%d", &choice);

switch(choice) {
    case 1:
        pattern1();
        break;
    case 2:
        pattern2();
        break;
    case 3:
        pattern3();
        break;
    case 4:
        pattern4();
        break;
    default:
        printf("Invalid Choice!\n");
}

return 0;
}
```

CODE SCREENSHOTS -

```
#include <stdio.h>

// Function for Pattern 1: Decreasing right-angled triangle
void pattern1() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {
        for(j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
}

// Function for Pattern 2: Decreasing natural-number pattern
void pattern2() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {
        for(j = 1; j <= i; j++) {
            printf("%d", j);
        }
        printf("\n");
    }
}

// Function for Pattern 3: Decreasing repeated-number pattern
void pattern3() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {
        for(j = 1; j <= i; j++) {
            printf("%d", i);
        }
        printf("\n");
    }
}

// Function for Pattern 4: Reverse-number triangle
void pattern4() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {
        for(j = i; j >= 1; j--) {
            printf("%d", j);
        }
        printf("\n");
    }
}
```

```
}

int main() {
    int choice;

    printf("--- MENU ---\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");
    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            pattern1();
            break;
        case 2:
            pattern2();
            break;
        case 3:
            pattern3();
            break;
        case 4:
            pattern4();
            break;
        default:
            printf("Invalid Choice!\n");
    }

    return 0;
}
```

OUTPUT SCREENSHOT :

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % nano 25q.c
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % gcc 25q.c -o 25q
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./25q
--- MENU ---
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice (1-4): 1
Enter number of rows: 5
*****
****
***
**
*
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./25q
--- MENU ---
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice (1-4): 2
Enter number of rows: 5
12345
1234
123
12
1
```

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./25q
```

```
--- MENU ---
```

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

Enter your choice (1-4): 3

Enter number of rows: 5

55555

4444

333

22

1

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./25q
```

```
--- MENU ---
```

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

Enter your choice (1-4): 4

Enter number of rows: 5

54321

4321

321

21

1

```
manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % █
```